

Fanconi Anemia Pathway Activation Converts Oncogene-Induced Senescence into a Pro-Tumorigenic State

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Abstract

DNA repair pathways are canonically tumor-suppressive, yet emerging evidence suggests their upregulation can promote cancer progression, a paradox whose mechanisms remain poorly understood. Here, we report that activation of the Fanconi Anemia (FA) DNA repair pathway drives malignant tumor progression by reprogramming cellular senescence from a tumor-suppressive to a pro-tumorigenic state. Through an unbiased genetic screen in *Drosophila*, we identified that loss of *Sras* (homologous to human *RCE1*) cooperates with oncogenic *Ras*^{V12} to induce slow-progressing yet highly malignant tumors. Bioinformatics analysis revealed marked upregulation of the FA pathway and Hippo signaling in these tumors. Genetic manipulation demonstrates that FA pathway activation is required for tumor growth and invasion. Mechanistically, we show that FA pathway upregulation enhances cellular senescence and triggers secretion of senescence-associated secretory phenotype (SASP) factors, particularly the cytokine Unpaired-1 (Upd1; IL-6 homolog), which in turn activates JAK-STAT signaling within tumor cells through an autocrine loop, thereby promoting tumor progression. Notably, p53 activation is dispensable for this pro-tumorigenic senescence program. FA pathway activation and its pro-tumorigenic function are conserved across multiple *Ras*^{V12}-driven malignant tumor models. Our findings reveal a previously unrecognized mechanism by which DNA repair machinery is co-opted to convert oncogene-induced senescence into a tumor-promoting state, with broad implications for understanding cancer progression and developing therapies targeting Ras-driven malignancies.

Key words: Fanconi Anemia pathway; Cellular senescence; SASP; RAS; *Sras*; Hippo pathway; JAK-STAT pathway; Cancer progression; *Drosophila*

Significance

The Fanconi Anemia (FA) DNA repair pathway is canonically recognized as a tumor suppressor. Here we reveal a paradoxical role where cancer cells co-opt the FA pathway to drive malignancy. We demonstrate that FA pathway activation converts oncogene-induced senescence from a tumor barrier into a pro-tumorigenic state through secretion of inflammatory cytokines that activate autocrine growth signaling. This mechanism operates across multiple *Ras*-driven tumor models, suggesting a general principle in cancer progression. These findings challenge the conventional view of DNA repair in oncology and identify the FA-senescence-SASP axis as a potential therapeutic vulnerability for RAS-mutant cancers.

Introduction

The Fanconi anemia (FA) pathway is a highly conserved DNA repair network that maintains genome integrity by resolving DNA interstrand crosslinks and stabilizing stalled replication forks. Named after Fanconi anemia, a rare inherited disorder caused by mutations in FA components, this syndrome is characterized by developmental defects, bone marrow failure, and a profound predisposition to hematological and solid malignancies^{1,2}. Canonically, the FA pathway functions as a tumor suppressor; its loss precipitates genomic instability and malignant transformation³. Paradoxically, however, emerging evidence indicates that hyperactivation or elevated expression of FA components correlates with tumor progression, metastasis, and therapeutic resistance across diverse cancer types⁴⁻⁹. For instance, upregulation of FANCD2 and FANCA in breast cancer is associated with poor prognosis⁸. While these observations suggest a context-dependent duality of the FA pathway in oncogenesis, the mechanistic basis by which FA pathway activation drives malignant progression remains elusive.

Cellular senescence represents a stable stress-induced state characterized by durable cell-cycle arrest and profound physiological remodeling. Oncogene-induced senescence (OIS), triggered by aberrant activation of oncogenic drivers such as *RAS*, serves as a potent intrinsic barrier to tumor initiation and early neoplastic expansion¹⁰⁻¹². Yet, senescent cells remain metabolically active, secreting a complex milieu of cytokines, chemokines, growth factors, and proteases known as the senescence-associated secretory phenotype (SASP). In established malignancies, SASP factors can remodel the tumor microenvironment to fuel proliferation, invasion, and metastatic dissemination¹³⁻¹⁷. Despite growing recognition of the dual roles of senescence in cancer, the molecular governing the transition from tumor-suppressive OIS to a pro-tumorigenic senescence program remain poorly defined.

RAS mutations occur in over 30% of human cancers, making them among the most frequent oncogenic drivers¹⁸. In *Drosophila*, overexpression of oncogenic *Ras*^{V12} in the wing imaginal disc induces benign, OIS-like growth. However, when combined with the loss of cell polarity genes, *Ras*^{V12} drives invasive malignant tumors that secrete SASP-equivalent factors, including the IL-6 homolog Unpaired (Upd) and matrix metalloproteinases, thereby activating JAK-STAT signaling to promote tumor progression¹⁹⁻²¹. Despite the centrality of *RAS* in oncology, effective therapeutic strategies remain limited, largely due to the complexity of the underlying signaling circuitry²². A fundamental question persists: how do *RAS*-driven tumor cells subvert intrinsic suppressive barriers and co-opt stress-response programs to fuel malignancy?

Drosophila melanogaster offers a genetically tractable *in vivo* platform to dissect

conserved mechanisms of tumorigenesis, enabling tissue-specific manipulation, mosaic clonal analysis, and the modeling of cooperative oncogenic interactions within an intact epithelium²³. Critically, core FA pathway components, including homologs of FANCD2 and FANCL, are functionally conserved in flies, establishing *Drosophila* as a robust system for investigating FA pathway dynamics during tumor progression²⁴⁻²⁶. Here, through an unbiased genetic screen, we identify that loss of *Sras*-the fly homolog of human *RCE1*, a CAAX prenyl protease essential for RAS post-translational processing, cooperates with oncogenic *Ras*^{V12} to drive highly malignant tumor progression. Transcriptomic profiling reveals a marked upregulation of the FA pathway and Hippo signaling in these tumors. Genetic experiments demonstrate that FA pathway activation is essential for tumor growth and invasion. Mechanistically, we show that FA pathway upregulation enhances cellular senescence and drives autocrine JAK-STAT activation via SASP-associated Upd1 secretion in a p53-independent manner, thereby converting OIS into a pro-tumorigenic state. These findings uncover a previously unrecognized mechanism by which DNA repair machinery is co-opted to promote *RAS*-driven malignancy, offering broad implications for cancer biology and therapeutic development.

Results

Loss of the RCE1 homolog *Sras* cooperates with *Ras*^{V12} to drive malignant progression

To identify novel genetic modifiers of *RAS*-driven tumor progression, we performed a large-scale ethyl methanesulfonate (EMS) mutagenesis screen using a previously established screening platform²⁷⁻³⁰. In this system, somatic mutations were introduced into *Ras*^{V12}-expressing cells within the eye-antennal imaginal disc epithelium. Mosaic clones were labeled with green fluorescent protein (GFP) via the MARCM (Mosaic Analysis with a Repressible Cell Marker) technique³¹, enabling the unambiguous *in vivo* identification of tumor progression and metastatic phenotypes (Fig. 1A). From this screen, we isolated two independent mutant alleles of a single gene named *Sras* (short for *severas*, CG4852). *Sras* is the *Drosophila* ortholog of human RCE1 (Ras converting enzyme 1), a membrane-bound endoplasmic reticulum protease that cleaves the C-terminal three amino acids of CAAX proteins³². Although the role of RCE1 in cancer remains incompletely defined, existing evidence links it to oncogenic *RAS* signaling. Notably, *RCE1* deficiency in mammals accelerates *KRAS*-induced myeloproliferative disease³³, suggesting that *RCE1* loss may paradoxically promote, rather than suppress, *RAS*-driven malignancy specific contexts. However, whether *Sras* functions in epithelial tumor progression had remained unexplored.

The two isolated *Sras* mutants, *Sras*^[1] and *Sras*^[2], harbor nonsense mutations at Q24 and

W140, respectively. Both mutations result in premature protein termination and disruption of the CPBP (CAAX proteases and bacteriocin-processing) functional domain (Fig. 1B). As the the Q24X mutation occurs earlier in the coding sequence, the *Sras*^[1] allele is predicted to produce a more severely truncated protein and was selected for all subsequent analyses. *Ras*^{V12}/*Sras*^{-/-} tumors exhibited a clear proliferative advantage over benign *Ras*^{V12}-only tumors, indicating that *Sras* loss significantly accelerates tumor progression (Fig. 1C). Notably, however, the kinetics of *Ras*^{V12}/*Sras*^{-/-} tumors were distinct from the rapidly lethal *Ras*^{V12}/*scrib*^{-/-} malignant model. While *Ras*^{V12}/*scrib*^{-/-} tumors form massive masses by day 10 after egg laying (AEL), causing larval death with darkened body color by day 16 (Fig. 1C). *Ras*^{V12}/*Sras*^{-/-} tumors required until day 13 to reach comparable sizes, yet affected larvae survived beyond day 16 (Fig. 1C). We therefore define this phenotype as “slow-progressing giant tumors”. Despite their relatively indolent growth kinetics, late-stage *Ras*^{V12}/*Sras*^{-/-} tumors (day 13-16) displayed hallmarks of high malignancy, including metastatic lesions in the Malpighian tubules and gut (Fig. 1D-D') and long invasive protrusions into adjacent brain lobes and ventral nerve cords (Fig. 1E). Collectively, these findings establish *Ras*^{V12}/*Sras*^{-/-} as a distinct cancer model characterized by delayed progression but high metastatic potential.

The Fanconi anemia pathway is hyperactivated and essential for *Ras*^{V12}/*Sras*^{-/-} tumor growth

To elucidate the molecular mechanism driving *Ras*^{V12}/*Sras*^{-/-} tumorigenesis, we performed microarray analysis comparing *Ras*^{V12} and *Ras*^{V12}/*Sras*^{-/-} tumors tissues. Gene Set Enrichment Analysis (GSEA) revealed that the Fanconi anemia (FA) pathway and Hippo signaling pathway were among the most significantly enriched pathways in *Ras*^{V12}/*Sras*^{-/-} tumors, a finding consistent across both *Sras* mutant alleles (Fig. 2A). Within the FA pathway, transcriptomic data showed significant upregulation of the *FANCI-Fancd2* complex, which constitutes the central axis of FA pathway activation, as well as *Rad51c* and *Rad51d*, which function downstream of the ID2 complex as Rad51 paralogs in homologous recombination (Fig. 2B)⁴.

To validate these findings *in vivo*, we employed a *Fancd2*-GFP transcriptional reporter³⁴. In *Ras*^{V12}/*Sras*^{-/-} tumors clones (labelled by RFP), we detected markedly elevated *Fancd2* transcription specifically within tumor cells (Fig. 2C). Since canonical FA pathway activation requires mono-ubiquitination of the FANCD2-FANCI complex to recruit downstream repair effectors^{35,36}, we assessed FANCD2 ubiquitination status. Co-immunoprecipitation (Co-IP) of FANCD2-GFP followed by ubiquitin blotting revealed a significant increase in FANCD2 mono-ubiquitination in *Ras*^{V12}/*Sras*^{-/-} tumors compared to controls (Fig. 2D). Thus, *Sras* loss drives robust FA pathway activation through a dual mechanism: transcriptional upregulation of

Fancd2 coupled with enhanced post-translational mono-ubiquitination of the FANCD2 protein.

Next, to determine the functional necessity of this activation, we used RNA interference (RNAi) to knock down key FA components (*Fancd2*, *Rad51C*, and *Rad51D*). Suppression of these factors yielded consistent and potent anti-tumor effects: tumor size was markedly reduced (Fig. 2E, E'), larval pupation rates significantly increased (Fig. S1A), and invasion into the ventral nerve cord (VNC) was effectively suppressed (Fig. S1B). Given the canonical role of the FA pathway in DNA damage repair, we examined intratumoral DNA damage levels using γ H2AX as a marker for double-strand breaks. Surprisingly, γ H2AX levels were significantly reduced within *Ras^{V12}/Sras^{-/-}* tumor clones compared to controls (Fig. S1C). This reduction was reversed upon knockdown of FA components (Fig. S1C, S1D) or overexpression of *Sras* (Fig. S1E), indicating that FA pathway activation in these tumors actively attenuates intratumoral DNA damage. Collectively, these data demonstrate that the FA pathway is transcriptionally and functionally hyperactivated in *Ras^{V12}/Sras^{-/-}* tumors, where it plays a critical role in promoting tumorigenesis and malignant progression.

FA pathway activation sustains Hippo/Yki signaling to promote tumorigenesis

Our microarray analysis further identified significant enrichment of the Hippo signaling pathway in *Ras^{V12}/Sras^{-/-}* tumors (Fig. 2A). Hierarchical clustering of Hippo-associated genes revealed broad transcriptional dysregulation, with marked upregulation of key downstream targets conserved across both *Sras* alleles (Fig. 3A). Immunostaining confirmed robust upregulation of Hippo pathway targets, including dMyc (*Drosophila* Myc), Wingless (Wg), and Expanded (*ex*, monitored by *ex-lacZ* reporter) (Fig. 3B-D).

Given the requirement for FA pathway activation drives this elevated Hippo/Yki transcriptional output. Remarkably, RNAi-mediated knockdown of FA pathway components significantly reduced Wg protein levels and suppressed *ex-lacZ* reporter activity within tumor clones (Fig. 3C, 3D). Thus, inhibiting the FA pathway is sufficient to attenuate the upregulation of multiple Hippo/Yki downstream targets in *Ras^{V12}/Sras^{-/-}* tumors. These findings suggest that the FA pathway may function upstream of, or in parallel with, Hippo signaling to sustain Yki-dependent transcription during tumorigenesis.

To further determine if Yki activation is functionally required for *Ras^{V12}/Sras^{-/-}* tumorigenesis, we genetically suppressed Yki activity using two approaches: overexpression of Warts (Wts), the core kinase that phosphorylates and inhibits Yki^{37,38}, and introduction of a heterozygous loss-of-function *yki* mutation. *wts* overexpression markedly reduced both the overall size and GFP-positive area of tumor clones, allowing a subset of larvae to survive to adulthood (Fig. 3E-E', S2A). Similarly, *yki* heterozygosity significantly decreased the GFP-

positive tumor area, though it did not profoundly reduce total disc size (Fig. 3E-E'). Together, these results demonstrate that dysregulation of Hippo signaling is a critical driver of *Ras*^{V12}/*Sras*^{-/-} progression and suggest that the FA pathway promotes tumor growth, at least in part, by sustaining Hippo pathway-dependent transcriptional output.

FA pathway activation drives a p53-independent senescence program

Having established that FA pathway activation promotes malignant growth and sustains Hippo/Yki signaling in *Ras*^{V12}/*Sras*^{-/-} tumors, we sought to characterize the cellular state underlying the unique “slow-progressing yet malignant” phenotype of these tumors. Staining for apoptotic markers (cleaved Dcp1 and cleaved Caspase-3) revealed low basal apoptosis in *Ras*^{V12}/*Sras*^{-/-} tumor clones (Fig. S3A-B), reminiscent of the apoptotic resistance characteristic of senescent cells^{39,40}. This prompted us to investigate whether these tumor cells acquire a senescence-like state.

We examined a panel of established senescence markers. SA- β -galactosidase activity, detected using the fluorogenic substrate SPiDER- β gal⁴¹, serves as a canonical readout of cellular senescence. While benign *Ras*^{V12} tumor clones displayed moderate SA- β -gal elevation consistent with oncogene-induced senescence (OIS)^{20,42}, this signal was dramatically amplified in *Ras*^{V12}/*Sras*^{-/-} tumor clones (Fig. 4A). Crucially, this amplification was attenuated by RNAi-mediated knockdown of FA pathway components (Fig. 4A). Parallel staining for Heterochromatin Protein 1 (HP1), a marker of senescence-associated heterochromatin foci, yielded a concordant pattern: HP1 was markedly elevated in *Ras*^{V12}/*Sras*^{-/-} tumors and suppressed upon FA inhibition (Fig. 4B). Together, these results indicate that FA pathway activation drives a robust senescence program in *Ras*^{V12}/*Sras*^{-/-} tumors.

Previous reports indicate that Ras-induced cell cycle arrest depends on p53 activation in both *Drosophila* and mice^{20,43}. Consistent with this, we observed strong upregulation of p53 and the G1 CDK inhibitor Dacapo (Dap; homolog of human p21/p27) in *Ras*^{V12}/*Sras*^{-/-} tumor clones (Fig. 4C, Fig. S3C), confirmed transcriptionally by a *dap-lacZ* reporter (Fig. S3D). Notably, both FA pathway knockdown and *wts* overexpression significantly reduced p53 levels in *Ras*^{V12}/*Sras*^{-/-} clones (Fig. 4B, Fig. S3E), placing p53-mediated arrest downstream of the FA-Hippo axis. However, direct functional testing using a dominant-negative p53 (*p53*^{DN}) revealed that p53 inactivation affected neither tumor size, nor Hippo target gene expression, nor the senescence status of *Ras*^{V12}/*Sras*^{-/-} clones (Fig. S3F-H). These results indicate that while p53 is upregulated, it is dispensable for maintaining Hippo-dependent transcription and the senescent state in this context. Instead, additional p53-independent mechanisms, driven by the FA pathway, sustain the pro-tumorigenic senescence program.

Senescence-associated Upd1 secretion activates autocrine JAK-STAT signaling to fuel *Ras*^{V12}/*Sras*^{-/-} tumor growth

Senescent cells typically secrete a spectrum of cytokines, chemokines, proteases known as the senescence-associated secretory phenotype (SASP). To determine if a SASP-like program is induced in *Ras*^{V12}/*Sras*^{-/-} tumors, we examined the expression of *Drosophila* homologs of human SASP factors^{16,44}. qRT-PCR analysis revealed significant upregulation of *wb*, *Pgr*, *PIG-F*, *upd1*, and *upd3* in *Ras*^{V12}/*Sras*^{-/-} tumors (Fig. 5A). Consistently, *Mmp1* protein levels and *upd1-lacZ* reporter activity reflecting the transcriptional level of *upd1* were strongly elevated in tumor clones (Fig. 5B,5C).

To assess the functional contribution of individual SASP factors, we performed genetic knockdowns. Depletion of *Mmp1* markedly reduced tumor clone area, suppressed tumor proliferation, and allowed complete larval pupation by day 10, with a subset of animals eclosing as adults (Fig. 5D, 5D', Fig.S4A, S4B). In contrast, *PIG-F* knockdown had no significant effect, likely due to insufficient RNAi efficiency (Fig. 5D, 5D'). We next focused on the Upd cytokines. At day 8, knockdown of *upd1* or *upd2* significantly reduced the GFP-positive tumor clone area, whereas *upd3* knockdown had no significant effect (Fig. 5E-5E''). By day 10, *upd1* depletion promoted complete pupation of all tumor-bearing animals, whereas *upd2* or *upd3* knockdown failed to alter tumor size or pupation rates (Fig. S4C-F). These data identify Upd1 as the predominant SASP factor driving *Ras*^{V12}/*Sras*^{-/-} tumor growth and developmental arrest.

Importantly, knockdown of FA pathway components substantially suppressed *upd1-lacZ* activity in *Ras*^{V12}/*Sras*^{-/-} tumors (Fig. 5C), as did overexpression of the Hippo kinase Wts (Fig. 5C, 5F). Conversely, expression of dominant-negative p53 had no effect on *upd1-lacZ* induction (Fig. 5F), confirming that SASP production is governed by the FA-Hippo axis independently of p53.

In *Drosophila*, Upd cytokines signal through the receptor Domeless (Dome), activating the JAK kinase Hopscotch (Hop) and promoting nuclear translocation of STAT92E⁴⁵⁻⁴⁷. Using a STAT-GFP reporter, we observed that STAT activation was predominantly confined to RFP-labeled tumor clones, with only weak activation in surrounding tissue (Fig. 5G), suggesting an autocrine mode of action. Consistent with this, overexpression of dominant-negative Domeless (Dome^{DN}) in *Ras*^{V12}/*Sras*^{-/-} tumor clones robustly reduced tumor clone area and disc size, increased pupation rate, and allowed a subset of animals to eclose into adults with rough-eye phenotypes (Fig. 5H-5H'', S4G, S4H). Collectively, these data imply that senescent tumor cells promote malignancy predominantly through autocrine Upd1/JAK-STAT signaling.

FA pathway engagement is a conserved feature of diverse *Ras*-driven polarity-deficient

tumors

Next, to assess whether the FA pathway plays a broader role in *Drosophila* tumorigenesis, we systematically analyzed FA pathway gene expression across multiple *Ras*^{V12}-driven tumor models using the Atlas of *Ras*-driven Tumors in *Drosophila* (ART-D) developed by our laboratory⁴⁸. FA pathway genes were found to be broadly upregulated across multiple *Ras*^{V12}-driven tumor models. Notably, in *Ras*^{V12}/*dlg*^{RNAi} model, FA pathway genes were strongly upregulated as early as day 5.5, indicating early activation during tumor initiation. In the *Ras*^{V12}/*scrib*^{-/-} model, FA expression exhibited a progressive, sustained increase from day 5.5 to day 13.5 (Fig. 6A), suggesting a stage-dependent involvement of the FA pathway activation in tumor progression.

Motivated by these expression patterns, we functionally validated the FA pathway's contribution in additional well-established polarity-deficient *Ras*-tumor models: *Ras*^{V12}/*scrib*^{-/-}, *Ras*^{V12}/*lgl*^{-/-}, and *Ras*^{V12}/*dlg*^{RNAi} 31,49,50. In the *Ras*^{V12}/*dlg*^{RNAi} model, *Rad51C* knockdown robustly suppressed tumor clone size and reduced eye-antennal disc area (Fig. 6B, 6B', S5A), demonstrating a requirement for FA activity in tumor growth. However, given the high basal pupation rate of *dlg* knockdown flies, FA knockdown did not significantly improve developmental progression (Fig. S5B). In the *Ras*^{V12}/*lgl*^{-/-} model, knockdown *Rad51C* or *Fancd2* significantly elevated larval pupation rates (Fig. S5C), indicating that FA inhibition alleviates systemic tumor-induced lethality. Paradoxically, however, FA inhibition failed to reduce local tumor area; *Rad51C* knockdown was even associated with an increase in disc area (Fig. S5D-S5F). These discordant results suggest that in the *lgl* context, the FA pathway primarily contributes to systemic lethality rather than driving local proliferation. Finally, in the *Ras*^{V12}/*scrib*^{-/-} model, *Rad51C* knockdown showed a trend toward increased pupation, though this did not reach statistical significance (Fig. S5G). Collectively, these findings indicate that while the FA pathway is broadly engaged across diverse *Ras*^{V12}-driven polarity-deficient tumors, its functional contribution-whether driving local growth or systemic burden-is highly context-dependent.

Discussion

In this study, we uncover a previously unrecognized role for the Fanconi anemia (FA) pathway as a central orchestrator of malignant progression in *Ras*^{V12}/*Sras*^{-/-} tumors. We demonstrate that aberrant FA pathway activation drives tumorigenesis through two distinct yet interconnected signaling axes: a cell-autonomous axis that suppresses the Hippo tumor suppressor pathway to fuel proliferation, and a senescence-dependent axis that leverages the senescence-associated secretory phenotype (SASP) to establish a self-reinforcing autocrine JAK-STAT loop.

Mechanistically, oncogenic transformation in *Sras*-deficient contexts promotes enhanced FANCD2 monoubiquitination and protein stabilization. This hyperactivated FA state simultaneously attenuates intratumoral DNA damage while triggering a robust senescence program. Crucially, rather than acting as a barrier, this FA-driven senescence facilitates the secretion of Upd cytokines, which activate JAK-STAT signaling within the tumor cells themselves. Together, these findings redefine the FA pathway not merely as a guardian of genome integrity, but as a pivotal node that co-opts stress-response programs to convert a potentially tumor-suppressive senescent state into a engine for malignant growth (Fig. 6C).

Our observation that FA pathway activation concurrently drives senescence and sustains tumor growth adds a critical dimension to the evolving understanding of oncogene-induced senescence (OIS). Classically, OIS is viewed as a cell-autonomous barrier wherein aberrant mitogenic signaling triggers replication stress and p53/p21-dependent arrest^{20,43}. However, emerging evidence suggests that senescent cells can paradoxically fuel progression via their secretome^{51,52}. Our work bridges these concepts by identifying the FA pathway as the upstream switch that enables this duality. We show that tumor cells exploit FA-mediated repair not only to survive oncogenic stress, consistent with reports of FANCD2 stabilizing replication forks⁵³, but also to actively maintain a pro-tumorigenic senescent state. Notably, the maintenance of this senescence and the associated SASP in *Ras*^{V12}/*Sras*^{-/-} tumors occurs independently of p53 function. This finding is particularly significant for human oncology, as it implies that in tumors harboring *TP53* mutations, where canonical OIS barriers are compromised, alternative FA-Hippo-driven mechanisms may still enforce a senescent-like state that inadvertently promotes malignancy via SASP. Furthermore, given that *Sras* (RCE1) loss alters Ras processing and dosage, and considering recent evidence linking *Ras* dosage to senescence fate⁵⁴, it is plausible that the FA pathway acts as a sensor of altered *Ras* signaling intensity, tuning the cellular response from arrest to aggressive expansion.

The functional coupling of FA pathway activation to Hippo pathway suppression represents an unrecognized regulatory axis in cancer biology. While the Hippo pathway's role in restraining YAP/TAZ (Yki in *Drosophila*) to limit organ growth is well-established^{55,56}, its integration with DNA damage response networks has remained elusive. Our data provide genetic evidence that FA activity is required for sustaining Yki-dependent transcriptional output in *Drosophila* tumors, effectively positioning the FA pathway as a potential upstream regulator of Hippo signaling. Given that both FA pathway dysregulation and YAP/TAZ hyperactivation are frequently observed in diverse human epithelial cancers⁵⁷, determining the conservation of this crosstalk in mammals is a priority. If conserved, this axis offers a compelling rationale for

targeting the FA pathway to restore Hippo-mediated tumor suppression, particularly in contexts where direct YAP/TAZ inhibition has proven challenging.

Distinct from the canonical model where senescent stromal cells promote neighboring tumor growth via paracrine signaling⁵⁸⁻⁶², our study reveals that *Ras*^{V12}/*Sras*^{-/-} tumors utilize a predominantly autocrine SASP mechanism. Here, Upd cytokines, particularly Upd1, act within the tumor clone to create a self-reinforcing loop that amplifies JAK-STAT-driven proliferation. This mirrors observations in human malignancies where IL-6 and other cytokines function autocrinely to sustain survival and therapy resistance^{63,64}. The identification of the FA-Hippo-SASP axis suggests a hierarchical signaling cascade wherein upstream inhibition of the FA pathway could simultaneously disrupt three pro-tumorigenic pillars: (1) restore Hippo-mediated growth control, (2) attenuate the pro-tumorigenic senescence program, and (3) abolish SASP-driven JAK-STAT activation. With small-molecule inhibitors targeting FANCD2 ubiquitination, RAD51-mediated recombination, and JAK kinases already in clinical or preclinical development^{8,65}, combinatorial strategies targeting this network offer a promising therapeutic avenue for *RAS*-driven epithelial malignancies that remain refractory to conventional therapies.

Still, several important questions remain. First, while our findings in *Drosophila* provide a robust genetic framework, validating the conservation of the FA-Hippo-SASP axis in mammalian epithelial tumors is essential. Second, while we identified Upd cytokines as key mediators, comprehensive proteomic profiling of the SASP in FA-hyperactive tumors may reveal additional effectors contributing to the tumor microenvironment. Third, the precise molecular intermediates linking FA pathway components to Hippo kinase regulation warrant detailed biochemical investigation. Finally, while our genetic data strongly support the therapeutic potential of targeting this axis, dedicated pharmacological studies in relevant *in vivo* models will be required to optimize dosing and evaluate efficacy in combination with existing standard-of-care treatments. Nevertheless, by elucidating how DNA repair machinery is co-opted to subvert senescence and drive malignancy, this study opens avenues for understanding and treating aggressive, *RAS*-driven cancers.

Materials and Methods

Fly Stocks and Genetics

All experiments were conducted at 25°C on standard *Drosophila* medium containing cornmeal, glucose, wheat germ, agar, and yeast. The composition of the standard food medium is as follows: 50 g cornmeal, 9 g agar, 24.5 g dry yeast, 7.25 g white sugar, 30 g brown sugar, 4.4 ml

propionic acid, 12.5 ml ethanol, and 1.25 g nipagin per liter of medium. Mitotic clones labeled by fluorescence in larval eye-antennal imaginal discs were generated by the MARCM technique using the 79E Tester (*ey-FLP1; Act>y+>Gal4, UAS-GFP; Tub-Gal80, FRT79E*) or 79E RFP Tester (*ey-FLP1; Act>y+>Gal4, UAS-RFP; Tub-Gal80, FRT79E*). All *Drosophila* lines used in this study were listed and described in Supplementary Table 1.

Immunostaining

Third-instar larval eye-antennal imaginal discs were dissected in cold PBS and fixed with 4% paraformaldehyde for 15 min at room temperature (RT). After washing three times with PBS containing 0.1% Triton X-100 (PBST), the samples were then blocked in PBS with 5% normal goat serum for 1 h at RT and incubated with primary antibodies overnight at 4 °C. After washing three times with PBST, all samples were incubated with secondary antibodies away from light for 2 h at RT. The samples were then dissected and mounted with a DAPI-containing medium and imaged using Zeiss fluorescence microscope. A complete list of antibodies is provided in Supplementary Table 2.

SA- β -gal staining

SA- β -gal activity was monitored by SPiDER- β Gal (SG02, Dojindo). Third-instar larvae were dissected in cold PBS and fixed with 4% paraformaldehyde for 5 min on ice and for 20 min at RT. After washing three times with PBS, the imaginal discs were incubated in SPiDER- β Gal working solution (SPiDER- β Gal dimethyl sulfoxide stock solution diluted to 1:1000 in fivefold diluted pH 6.0 McIlvaine buffer with distilled water) for 2 hours at 37°C and then washed three times with PBS. After mounting as described above, images were taken using Zeiss fluorescence microscope.

Clone size measurement

Clone size measurement was conducted by ImageJ software. Total disc size was measured by drawing the outline according to the DAPI channel. GFP positive area was obtained by setting threshold and then measured. Data was then imported to Microsoft Excel software and statistically analyzed through GraphPad Prism software.

RNA isolation and RT-qPCR

Drosophila third-instar larvae were washed twice with sterile dPBS and dissected in pre-cooled dPBS. The eye-antennal discs were collected and subsequently cryopreserved with liquid nitrogen in -80 °C for later use. Total RNA was extracted using TRIzol (Ambion), and cDNA was obtained by reverse transcription kit according to the manufacturer's instructions. Quantitative PCR was performed with SYBR Green Real-time PCR Master Mix to determine

the relative abundance of mRNA. Expression levels were normalized to *rp49* as a housekeeping gene. All primer sequences used in this study are listed in Supplementary Table 3.

Co-immunoprecipitation and Detection of Fancd2 Ubiquitination

To assess ubiquitination of Fancd2 in the tumor context, eye-antennal imaginal discs were dissected from third-instar larvae expressing *Ras^{V12}/Fancd2-GFP*. Tissues were lysed in IP lysis buffer supplemented with protease inhibitors. Clarified lysates were subjected to immunoprecipitation using an anti-ubiquitin antibody overnight at 4°C, followed by incubation with protein A/G agarose beads. Precipitated complexes were washed, eluted by boiling in SDS sample buffer, and resolved by SDS-PAGE. Ubiquitinated FANCD2-GFP was detected by immunoblotting with an anti-GFP antibody and signals were visualized by enhanced chemiluminescence (ECL).

Graphics

Fluorescence images were acquired using a Zeiss Axio Observer with ApoTome.2. Images were processed with Fiji/ImageJ software (<https://fiji.sc>) for image merging and resizing. Schematic panels in Fig. 1A, 1E, and 6C were created with BioRender.com (www.biorender.com).

Microarray Gene Expression Analysis

Microarray gene expression analysis was performed using *Drosophila* eye-antennal imaginal discs with tumors caused by *Ras^{V12}/Sras^{-/-}*. *Ras^{V12}* eye discs were used as a negative control, and *Ras^{V12}/scrib^{-/-}* tumors were used as a positive control for tumors with overgrowth and metastasis. RNA was isolated from tumors using the Qiagen RNeasy Mini Kit according to the manufacturer's protocol. Microarray analysis was performed by the Yale Center for Genome Analysis (YCGA) using an Affymetrix GeneChip *Drosophila* Genome 2.0 Array. Genes whose expression was more than 1.5-fold higher in *Ras^{V12}/Sras^{-/-}* eye discs than in *Ras^{V12}* eye discs were defined as upregulated genes using the limma package⁶⁶. Functional annotation of these upregulated genes was performed using clusterProfiler with the org.Dm.eg.db database and the DAVID online tool to identify enriched pathways⁶⁷.

Pan-Tumor Evaluation of FA Pathway via ART-D

To systematically evaluate the role of FA pathway across multiple *Ras^{V12}*-driven malignant tumors, we utilized the newly developed *Drosophila* tumor atlas, ART-D. This resource encompasses ten genetically defined *Ras^{V12}*-driven models spanning the initiation, promotion, and progression stages of tumorigenesis (5.5, 9.5, and 13.5 days AEL). The normalized gene

count matrix underwent $\log_2(\text{CPM}+1)$ transformation, and the heatmap visualization of FA pathway genes were subsequently row-scaled using Z-scores. The FA pathway gene list was accessible via Supplementary Table 4.

Data and Code Availability

The raw microarray data generated in this study are available in the Gene Expression Omnibus (GEO) repository under accession number XXXX. The analysis code is accessible via GitHub at <https://github.com/GYF-pixel/Ras-Sras-Malignant-Tumor>.

Quantification and Statistical Analysis

For each experiment, at least 5 discs per genotype were analyzed across a minimum of three independent biological replicates. Data are presented as mean $\pm$ SEM. Statistical comparisons between two groups were performed using Student's t-test, and comparisons among multiple groups were performed using one-way ANOVA followed by Tukey's post hoc test. All statistical analyses were conducted using GraphPad Prism 9.0. A P -value < 0.05 was considered statistically significant.

Conflict of interest

The authors declare no competing interests.

Acknowledgements

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Author contributions

X.M. conceived and supervised the project; X.M., S.S., W.X., and Y.Q. designed the experiments and analyzed the data; J.L. performed the initial genetic screen and mapping; Y.Q. conducted the majority of experiments with support from J.M., Y.G., X.L., and W.L.; Y.Q., X.M., and S.S. wrote the manuscript with input from all co-authors.

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Figures and Figure Legends

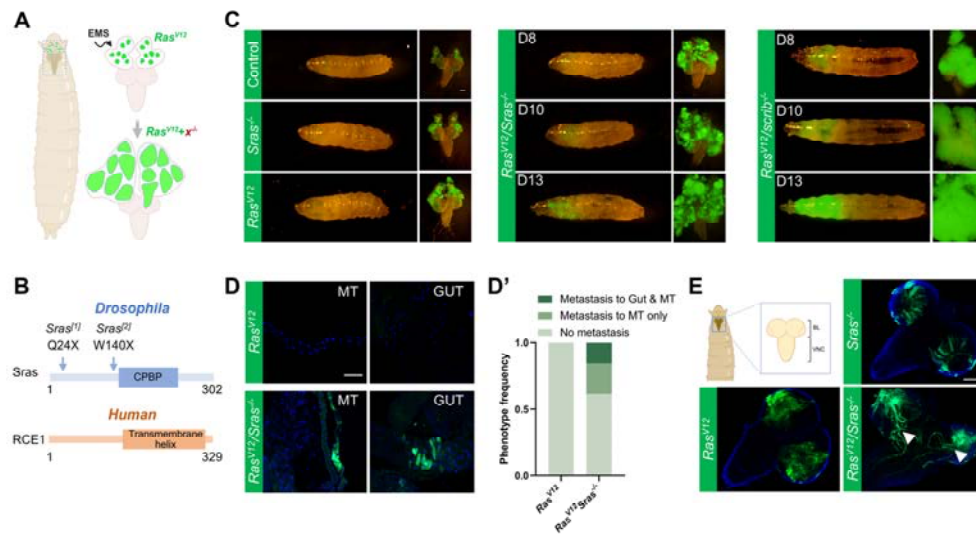


Fig 1. Loss of the *Sras* cooperates with Ras^{V12} to drive malignant progression.

(A) Schematic of the screening strategy used to identify mutations that drive the transformation of Ras^{V12} -induced benign tumors to malignancy. GFP-labeled tumor clones with EMS induced mutations were generated with MARCM technique in *Drosophila* eye-antennal discs.

(B) Schematic representation of the *Sras* protein domain structure, indicating the positions of the two nonsense mutations (Q24X and W140X).

(C) Representative image of whole larva (left) and cephalic complex (right) of control and tumors at the indicated timepoints. Scale bar, 100 μ m.

(D) Representative images showing tumor metastasis of Ras^{V12} and $Ras^{V12}/Sras^{-/-}$ larvae. MT, Malpighian tubule. (D') Quantification of tumor metastasis rates for the indicated genotypes (n = 10, 13). Scale bar, 100 μ m.

(E) Representative images of brain lobes (BL) and ventral nerve cords (VNC) from larvae of the indicated genotypes. Arrowheads indicate $Ras^{V12}/Sras^{-/-}$ tumor cells invading the VNC. Scale bar, 100 μ m.

Figure 2

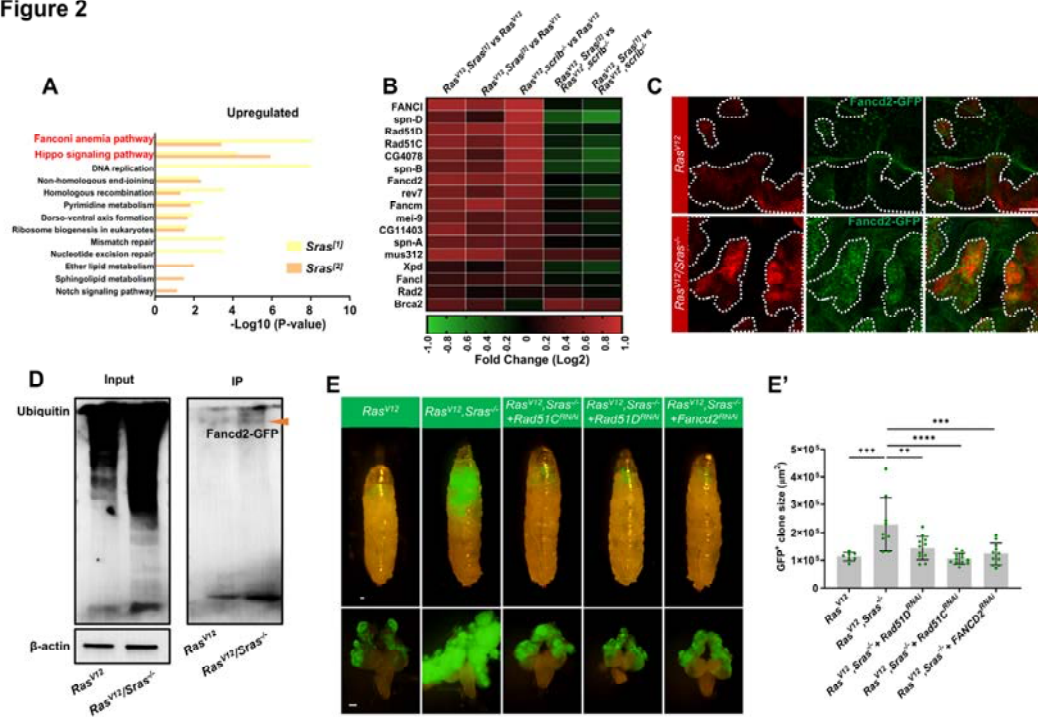


Fig 2. The FA pathway is hyperactivated and essential for *Ras^{V12}/Sras^{-/-}* tumor growth.

(A) GSEA pathway enrichment analysis of microarray data identifying upregulated pathways in *Ras^{V12}/Sras^{-/-}* tumors. *Sras^[1]* and *Sras^[2]* represent two independent *Sras* loss-of-function alleles.

(B) Heatmap showing transcriptional upregulation of FA pathway genes across indicated tumor genotypes.

(C) Fancd2 transcription activity assessed by the *Fancd2-GFP* reporter in *Ras^{V12}* and *Ras^{V12}/Sras^{-/-}* tumors. Tumor clones are outlined in white and co-labeled with RFP. Scale bar, 50μm.

(D) Co-immunoprecipitation (Co-IP) analysis of Fancd2 mono-ubiquitination in indicated tumor genotypes. Arrowhead indicates the Fancd2-GFP fusion protein.

(E) Representative image of larva (upper) and isolated cephalic complexes (lower) of *Ras^{V12}/Sras^{-/-}* tumors with indicated FA pathway gene knockdowns. (E') Quantification of GFP positive clone size. n=7 to 12 for each indicated genotype. Scale bar, 100μm.

One-way ANOVA was used to test significance in (E'). Data are shown as mean ± SD. ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001.

A

Fold Change (Log2)

1.5
1.0
0.5
0
-0.5
-1.0
-1.5

lilur3
ex
fat
Ack
d
Me
crb
Fbx07
CG10933
Pdx-1
Cdk
Cdk
wts
Gst
vd
jla
Hpk
CG4788
mits
Tao
hpyr
hpyr
Pis2A-29B
Rasaf
hoo
hoo
sav
14-3-3epitoin
Brip
xip
CG9133
Zyk
echinoid
warts
Shim
RafA
RhoGEF
14-3-3zeta
rmb
Shim
14-3-3zeta
RG37575

B

Control Ras^{V12} Ras^{V12}, Sras^{-/-}

dMyc

C

Ras^{V12} Ras^{V12}, Sras^{-/-} Ras^{V12}, Sras^{-/-} + Fancd2^{RNAi} Ras^{V12}, Sras^{-/-} + Rad51D^{RNAi}

ex-lacZ

D

Control Ras^{V12} Ras^{V12}, Sras^{-/-} Ras^{V12}, Sras^{-/-} + Fancd2^{RNAi} Ras^{V12}, Sras^{-/-} + Rad51C^{RNAi} Ras^{V12}, Sras^{-/-} + Rad51D^{RNAi}

Wg

E

Ras^{V12}, Sras^{-/-} Ras^{V12}, Sras^{-/-} + Wts Ras^{V12}, Sras^{-/-} + Yki1

E'

GFP positive area (μm²)

8x10⁵
6x10⁵
4x10⁵
2x10⁵
0

Ras^{V12}, Sras^{-/-} Ras^{V12}, Sras^{-/-} + Wts Ras^{V12}, Sras^{-/-} + Yki1

(A) Heatmap showing transcriptional upregulation of Hippo pathway across indicated tumor genotypes.

(C) Representative immunostaining images showing *ex-lacZ* reporter activity of indicated tumor genotypes. Scale bar, 50 μ m.

(E) Representative images of eye-antennal discs of indicated tumor genotypes. Quantification of GFP positive clone size is shown in E'. n=6 to 8 for each indicated genotype. Scale bar, 200 μ m.

Unpaired *t* test was used in (E'). Data are shown as mean $\pm$ SD. **P* < 0.05, *****P* < 0.0001.

Figure 4

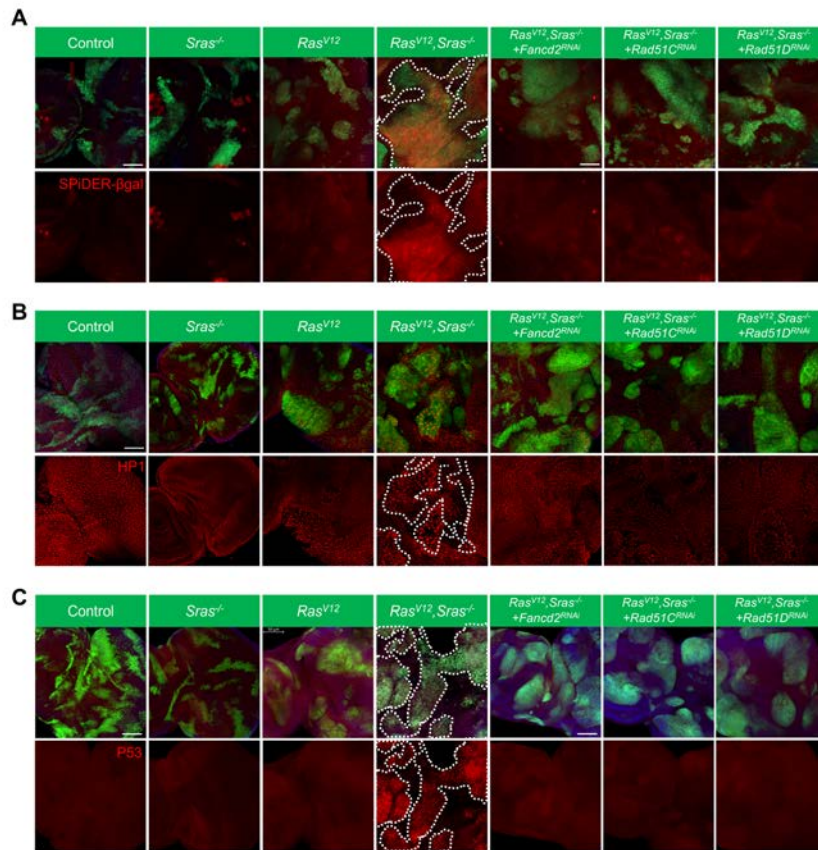


Fig 4. *Ras*^{V12}/*Sras*^{-/-} tumors induce FA pathway-dependent senescence.

(A) SPiDER-βgal staining for SA-β-gal activity in the indicated genotypes. Scale bar, 50μm.

(B) Immunostaining for HP1 in the indicated genotypes. Scale bar, 50μm.

(C) Immunostaining for p53 in the indicated genotypes. Scale bar, 50μm.

The white outlines indicate tumor clones labeled with GFP in eye-antennal discs.

Figure 5

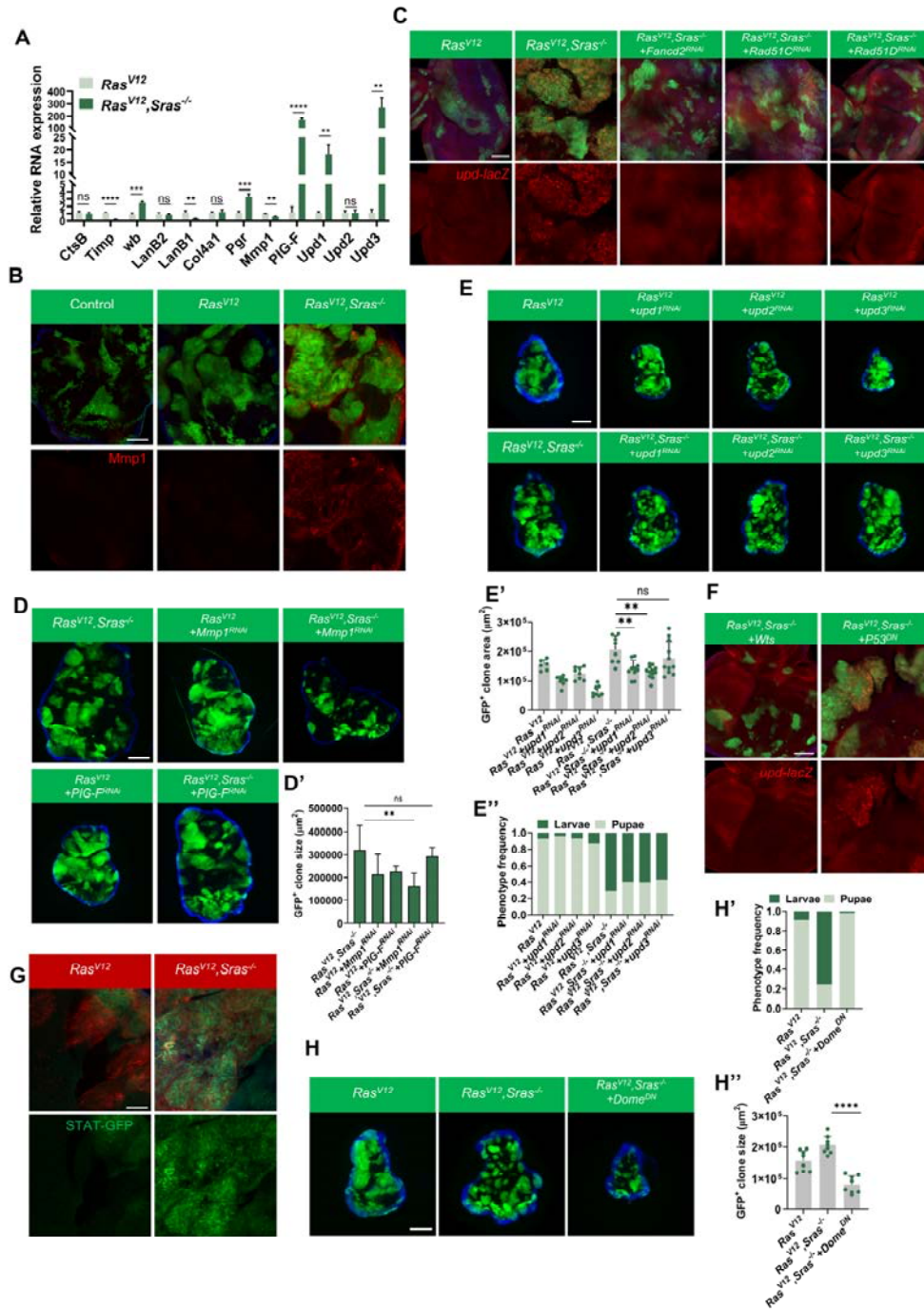


Fig 5. Senescence-associated Upd1 secretion activates autocrine JAK-STAT signaling to fuel *Ras^{V12}/Sras^{-/-}* tumor growth.

(A) qPCR analysis of SASP factor transcript levels in *Ras^{V12}* and *Ras^{V12}/Sras^{-/-}* tumors.

(B) Immunostaining for MMP1 in the indicated genotypes. Scale bar, 50µm.

(C) Detection of the *upd1-lacZ* reporter, which reflects *upd1* transcriptional activity, in the indicated genotypes. Scale bar, 50µm.

(D) Representative image of eye-antennal discs in the indicated genotypes. Quantification of GFP positive clone size is shown in D'. Scale bar, 100µm.

(E) Representative image of eye-antennal discs of *Ras^{V12}* or *Ras^{V12}/Sras^{-/-}* tumors with Upd1/2/3

Fanconi Anemia Pathway Activation Converts Oncogene-Induced Senescence into a Pro-Tumorigenic State

Yue Qiu, Jingjie Mu, Yifan Guo, Jin-Yu Lu, Xiaoqin Li, Wenahn Liu, Wenyan Xu, Sha Song, Xianjue Ma

Supplementary information contains the following sections:

- Supplementary Fig.S1-Fig.S5 and figure legends.
- Supplementary Table 1 *Drosophila* strains and genetic tools used in this study.
- Supplementary Table 2 Primary and secondary antibodies used in this study.
- Supplementary Table 3 Oligonucleotide primers used in this study.
- Supplementary Table 4 The FA pathway gene list.

Supplementary Figures and Figure Legends

Figure S1

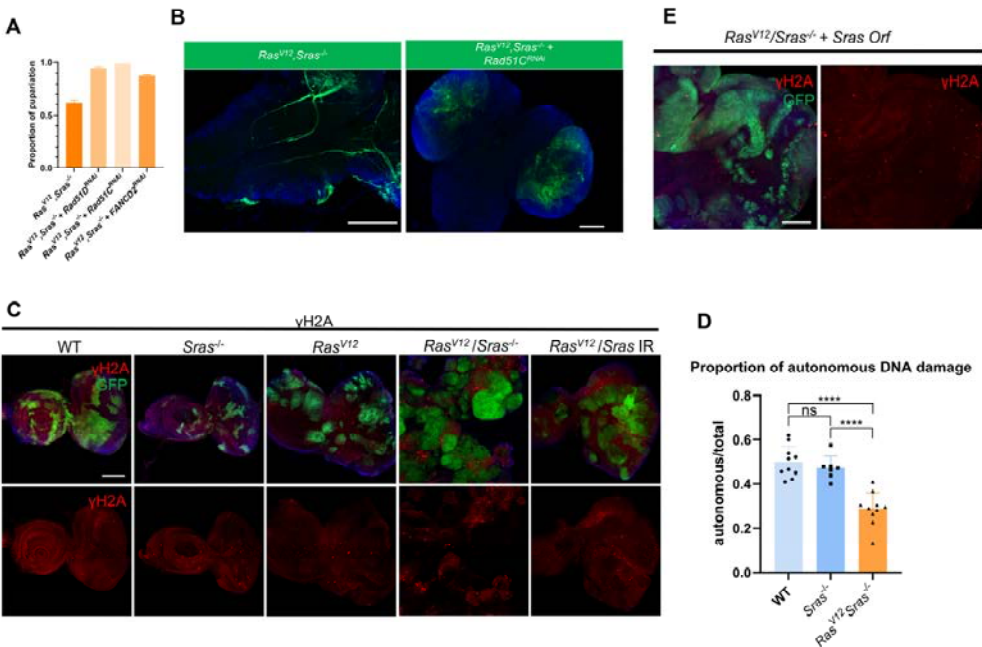


Fig S1. Low levels of DNA damage detected in *Ras^{V12}/Sras^{-/-}* tumors.

- (A) Larval pupation rates for the indicated genotypes.
- (B) Representative image of brain lobes (BL) and ventral nerve cords (VNC) harboring *Ras^{V12}/Sras^{-/-}* tumors, compared with tumors with *Rad51C* knockdown. Scale bar, 100μm.

- (C) Immunostaining of γ H2AX in eye-antennal imaginal discs of indicated genotypes, including $Ras^{V12}/Sras^{-/-}$ tumors with FA pathway gene knockdowns. Scale bar, 100 μ m.
- (D) Quantification of the proportion of γ H2AX-positive cells reflecting autonomous DNA damage within tumor clones of indicated genotypes.
- (E) Immunostaining of γ H2AX in $Ras^{V12}/Sras^{-/-}$ tumors with or without *Sras* overexpression. Scale bar, 50 μ m.

Figure S2

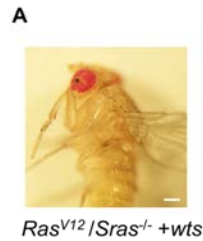


Fig S2. Genetic inhibition of Yki activity rescues $Ras^{V12}/Sras^{-/-}$ larval lethality.

- (A) Representative image of adult fly bearing $Ras^{V12}/Sras^{-/-}$ tumors with *wts* overexpression. Scale bar, 200 μ m.

Figure S3

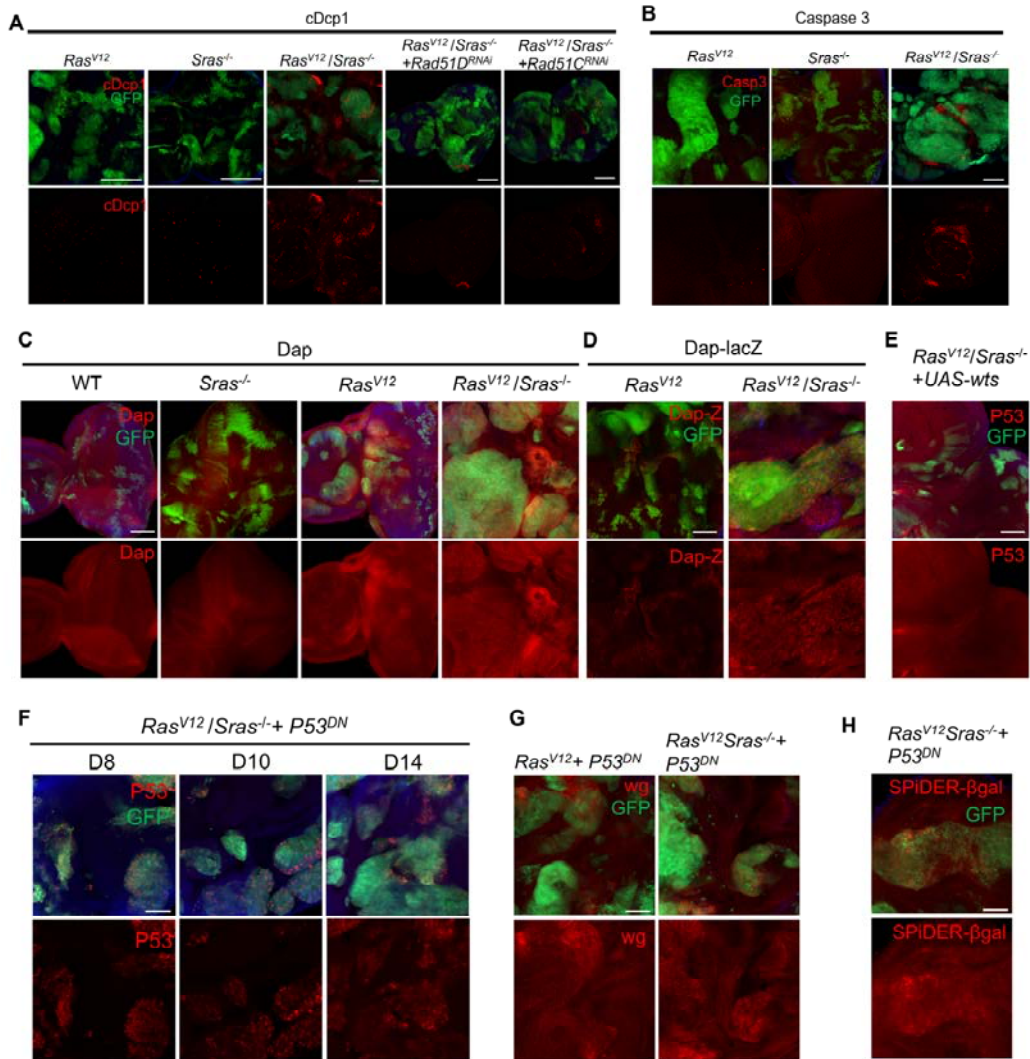


Fig S3. p53 activation is dispensable for senescence induction and tumorigenesis in *Ras^{V12}/Sras^{-/-}* tumors.

(A) Immunostaining for cleaved Dcp1 (cDcp1) in control, *Ras^{V12}/Sras^{-/-}* tumors and tumors with FA pathway gene knockdowns. Scale bar, 100µm.

(B) Immunostaining for Caspase 3 (Casp3) in control and tumor groups. Scale bar, 100µm.

(C) Immunostaining for Dacapo (Dap) in control and tumor groups. Scale bar, 50µm.

(D) Immunostaining for *dap-lacZ* in *Ras^{V12}* and *Ras^{V12}/Sras^{-/-}* tumors. Scale bar, 50µm.

(E) Immunostaining for p53 in *Ras^{V12}/Sras^{-/-}* tumors with *wt*s overexpression. Scale bar, 50µm.

(F) Immunostaining for p53 in *Ras^{V12}/Sras^{-/-}* tumors overexpressing dominant negative form of p53 (*p53^{DN}*) at Day 8, 10 and 14. Scale bar, 50µm.

(G) Immunostaining for Wingless (Wg) in *Ras^{V12}* or *Ras^{V12}/Sras^{-/-}* tumors overexpressing *p53^{DN}*. Scale bar, 50µm.

(H) Detection of SA-β-gal by SPiDER-βgal staining in *Ras^{V12}/Sras^{-/-}* tumors overexpressing *p53^{DN}*. Scale bar, 50µm.

Figure S4

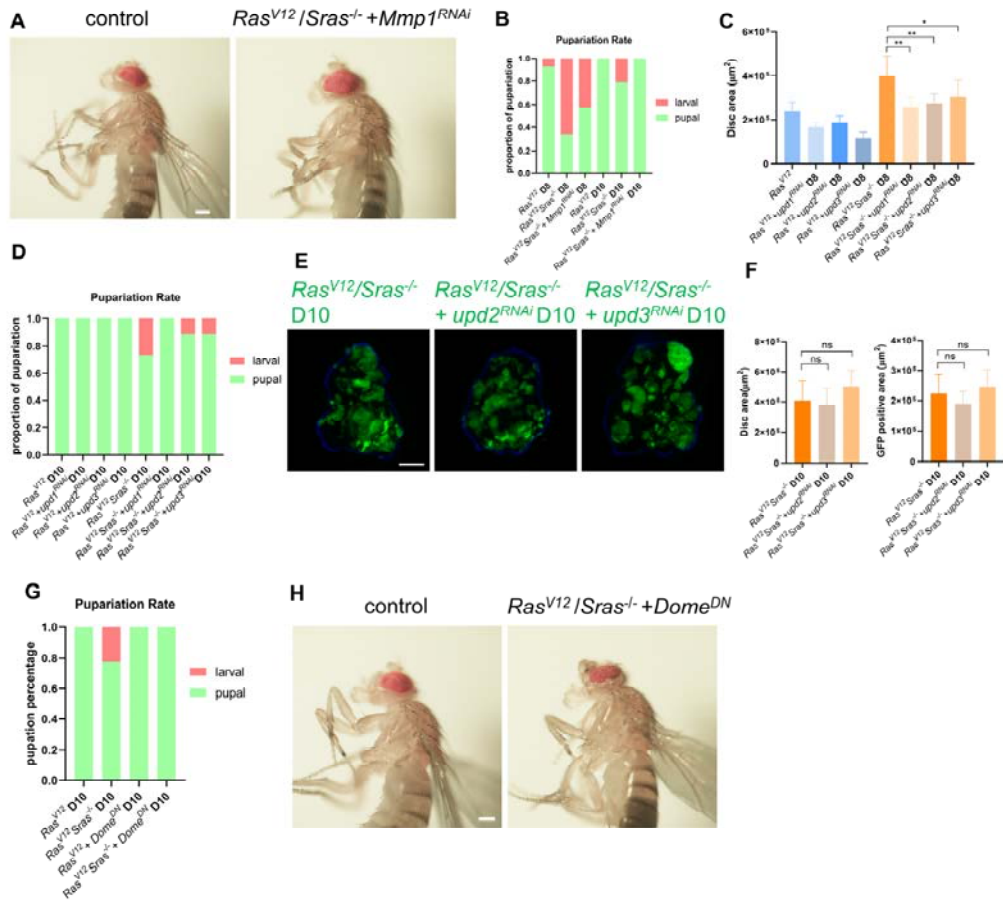


Fig S4. Knockdown of SASP factors or inhibition of JAK-STAT signaling suppresses the progression of *Ras^{V12}/Sras^{-/-}* tumors.

(A) Representative image of adult fly bearing *Ras^{V12}/Sras^{-/-}* tumors with *Mmp1* knockdown. Scale bar, 200 μm .

(B) Larval pupation rate of *Ras^{V12}/Sras^{-/-}* tumors with *Mmp1* knockdown at day 8 and 10.

(C) Quantification of total disc area of *Ras^{V12}* or *Ras^{V12}/Sras^{-/-}* tumors with *upd1*, *upd2*, or *upd3* knockdown at day 8. n = 6 to 12 for each indicated genotype.

(D) Larval pupation rate of *Ras^{V12}* or *Ras^{V12}/Sras^{-/-}* tumors with *upd1*, *upd2*, or *upd3* knockdown at day 10.

(E) Representative image of eye-antennal discs of *Ras^{V12}* or *Ras^{V12}/Sras^{-/-}* tumors with *upd2* or *Upd3* knockdown at day 10. Image of *Ras^{V12}/Sras^{-/-}* tumor with *upd1* knockdown is not shown due to complete pupation at day 10. Scale bar, 200 μm .

(F) Quantification of total disc area and GFP positive clone size of *Ras^{V12}/Sras^{-/-}* tumors with *upd2* or *upd3* knockdown at day 10. n=10 to 13 for each indicated genotype.

(G) Larval pupation rate of *Ras^{V12}* or *Ras^{V12}/Sras^{-/-}* tumors overexpressing *Dome^{DN}* at day 10.

(H) Representative image of adult fly bearing *Ras^{V12}/Sras^{-/-}* tumors overexpressing *Dome^{DN}*. Scale bar, 200 μm .

Unpaired *t* test was used in (C) and (F). Data are shown as mean $\pm$ SD. ns, no significance, **P* < 0.05, ***P* < 0.01.

Figure S5

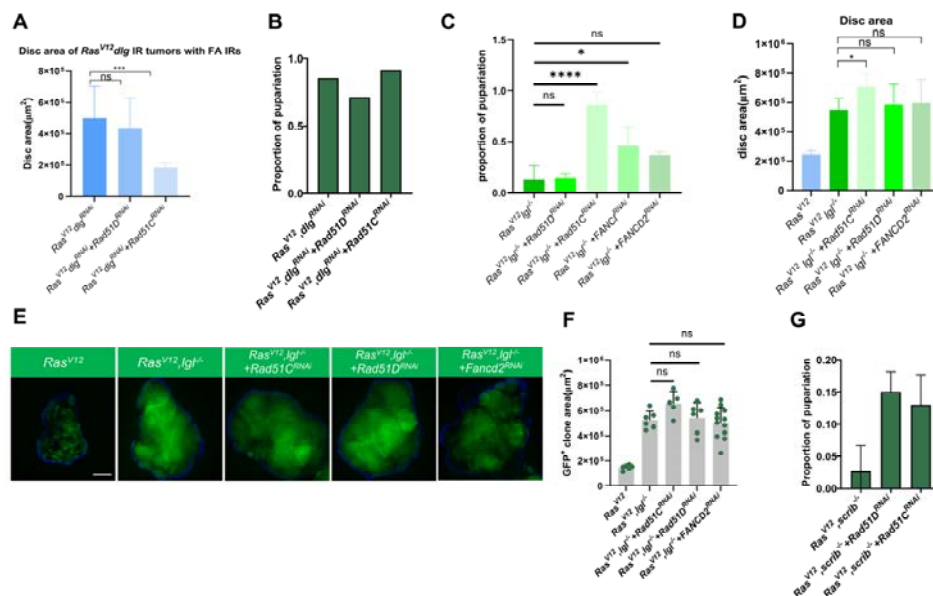


Fig S5. Differential dependency on the FA pathway between *Ras^{V12}/dlg^{RNAi}* and *Ras^{V12}/lgl^{-/-}* tumors.

(A) Quantification of total disc area in *Ras^{V12}/dlg^{RNAi}* tumors following knockdown of *Rad51D* or *Rad51C*. *n* = 4 to 10 for each indicated genotype.

(B) Larval pupation rate of *Ras^{V12}/dlg^{RNAi}* tumors with *Rad51D* or *Rad51C* knockdown.

(C) Larval pupation rate of *Ras^{V12}/lgl^{-/-}* tumors following knockdown of indicated FA pathway genes.

(D-F) Quantification of total disc area (D) and GFP positive clone size (F) of *Ras^{V12}/lgl^{-/-}* tumors with FA pathway gene knockdowns. *n* = 5 to 12 for each indicated genotype. Representative image of eye-antennal discs of each genotype is shown in (E). Scale bar, 200μm.

(G) Larval pupation rate of *Ras^{V12}/scrib^{-/-}* tumors following *Rad51D* or *Rad51C* knockdown.

Unpaired *t* test was used in (A), (D), and (F). Data are shown as mean $\pm$ SD. ns, no significance, **P* < 0.05, ****P* < 0.001.

Supplementary Table 1 *Drosophila* strains and genetic tools used in this study.

Drosophila line	Source	Catalog number
ex ^{e1} (ex-lacZ)	BDSC	Cat# 44249
UAS-dap.EGFP	BDSC	Cat# 83337
UAS-EGFP.dap	BDSC	Cat# 83338
Fancd2 ^{K595R} /TM3	BDSC	Cat# 94793
Ubi-Fancd2.EGFP	BDSC	Cat# 600616
Ubi-Fancd2 ^{K595R} .EGFP	BDSC	Cat# 600617
UAS-Stat92e ^{RNAi} /TM3	VDRC	Cat# 43867
UAS-hop ^{RNAi} /CyO	VDRC	Cat# 40037
UAS-Timp ^{RNAi}	VDRC	Cat# 15372
UAS-Col4a1 ^{RNAi}	VDRC	Cat# 28369
UAS-Sras ^{RNAi}	BDSC	Cat# 65173
UAS-Sras-shRNA	VDRC	Cat# 330678
UAS-FANCD2 ^{RNAi#1}	VDRC	Cat# 29518
UAS-FANCD2 ^{RNAi#2}	VDRC	Cat# 45433
UAS-FANCI ^{RNAi#1}	VDRC	Cat# 24655
UAS-Rad51C ^{RNAi}	VDRC	Cat# 1494
UAS-Rad51D ^{RNAi}	VDRC	Cat# 10758
UAS-Rad51D-ORF	FlyORF	Cat# F002360
UAS-Sras-ORF	FlyORF	Cat# F004475
UAS-FANCD2 ^{RNAi#3}	THFC	Cat# TH02453.N
UAS-FANCD2 ^{RNAi#4}	THFC	Cat# TH201500414.S
UAS-FANCI ^{RNAi#2}	THFC	Cat# THU3594
UAS-spn-D ^{RNAi#1}	BDSC	Cat# 67355
UAS-spn-D ^{RNAi#2}	VDRC	Cat# 25026
UAS-spn-D ^{RNAi#3}	THFC	Cat# TH05021.N
UAS-spn-A ^{RNAi#1}	BDSC	Cat# 31199
UAS-spn-A ^{RNAi#2}	BDSC	Cat# 51926
UAS-spn-A ^{RNAi#3}	VDRC	Cat# 14021

UAS-spn-A ^{RNAi#4}	THFC	Cat# TH02208.N
UAS-spn-A ^{RNAi#5}	THFC	Cat# TH201500859.S
Dap-lacZ	BDSC	Cat# 10406
RFP-HP1	BDSC	Cat# 30562
UAS-upd1 ^{RNAi}	VDRC	Cat# 3282
UAS-upd2 ^{RNAi}	VDRC	Cat# 330691
UAS-upd3 ^{RNAi}	VDRC	Cat# 27134
UAS-mmp1 ^{RNAi}	VDRC	Cat# 330331
UAS-dlg ^{RNAi}	VDRC	Cat# 41136
Diap1j5C8 (diap1-lacZ)	BDSC	Cat# 12093
UAS-bsk ^{DN}	BDSC	Cat# 6409
UAS-Ras ^{V12}	Gift from Tian Xu	(Pagliarini RA, et al., 2003, Science)
UAS-P53 ^{DN}	Gift from Tian Xu	
yki ^{B5}	Gift from DuoJia Pan	FBal0194179
UAS-hpo	Gift from Shian Wu	(Wu, et al., 2003, Cell)
UAS-wts	Gift from Shian Wu	(Wu, et al., 2003, Cell)
UAS-dome ^[ΔCYT1.1]	Gift from James Hombria	Andalusian Center
STAT-GFP	Gift from Erika Bach	New York University

BDSC: Bloomington Drosophila Stock Center; VDRC: Vienna Drosophila Resource Center; THFC: TsingHua Fly Center.

Supplementary Table 2 Primary and secondary antibodies used in this study.

Antibody	Producer	Cat no.
		Cat# 3A6B4
Mmp1	DSHB	Cat# 3B8D12
		Cat# 5H7B11
Dap	DSHB	Cat# NP1
β-galactosidase	DSHB	Cat# 40-1a
P53	DSHB	Cat# 25F4-s

P53	DSHB	Cat# 7A4
HP1	DSHB	Cat# C1A9
γ h2AX	DSHB	Cat# UNC93-5.2.1
Wg	DSHB	Cat# 4D4
Dcp-1	CST	Cat# 9578
Caspase-3	CST	Cat# 9664
Ubiquitin	CST	Cat# 3933S
PH3	CST	Cat# 9701
dMyc		
Goat anti-Mouse IgG (H+L) Alexa Fluor TM Plus 555	Invitrogen	Cat# A32727
Goat anti-Rabbit IgG (H+L) Alexa Fluor TM Plus 647	Invitrogen	Cat# A32733
Goat anti-Rabbit IgG (H+L) Alexa Fluor TM Plus 555	Invitrogen	Cat# A32732

CST: Cell Signaling Technology; DSHB: Developmental Studies Hybridoma Bank

Supplementary Table 3 Oligonucleotide primers used in this study.

RT-qPCR primer sequence (5'-3')	
Primer	5'-3' sequence
rp49-F	GCCGCTTCAAGGGACAGTAT
rp49-R	CGCTCGACATCTCTTGAGG
CtsB-mA-F	GAATGGAACCCGACCACCAT
CtsB-mA-R	CTCGTGCTGATAGACCGTCC
Timp-F	AACAGAGCGTCATGGCTTCA
Timp-R	ACCACTTGCATGTATCGGCA
EGFR-F	TGGATGAGCATATCGCGAGG
EGFR-R	TTGCTGACCGATGAGCTTGT
wb-F	TTCGAGCCTTTGCTTTTGCC
wb-R	TGGGACACTTACACCGTTTCG

LanB2-F	ATACGGATGGACCCGATTGC
LanB2-R	GATTCGCGTCGAAGAAGCAC
LanB1-F	GAAACGTATTCCGCCGAAGC
LanB1-R	GATTATTGGCCTCTCCGCGA
Col4a1-F	GGCGATATGGGACAGAAGGG
Col4a1-R	GTCCTGTAGATCCTGGGGGT
MMP1-F	CCTCGAGGCACCAATCTGTT
MMP1-R	TCGTCCAACCTGAAGACGGG
PTGER2-F	CCTGTGCTCGACCTGTTGAT
PTGER2-R	TCTTGCGCGCCAAAATGAAA
PIG-F-F	CTTCTGCGAGAAACCGGACT
PIG-F-R	GCGTAGAGGACATGCGTACA
upd1-F	TCGCAACTTCGACGAGAACA
upd1-R	ATTGAAGCTCTGCCTGGGAC
upd2-F	TCAATCCGTATCGCGGTCTG
upd2-R	AGAAGAGTCGCAGGTTGTGG
upd3-F	CATGTCCCAGTTTGCCCTCT
upd3-R	TCTCCCAGTCCAGATGCGTA

Supplementary Table 4 The FA pathway gene list, related to Fig. 6A.

FA_pathway_ID
Top3alpha
Erccl
Mlh1
Rev1
Fanc1
mms4
FANCI
RPA3
Usp1
Fancd2
slx1
PolZ1
mus81
spn-D

Top3beta

mei-9

mei-41

Blm

PolH

PolI

Fancm

spn-A

Pms2

Uaf1

RPA2

RPA1

Xrcc2

Rad51D

Rtel1

spn-B

PolZ2

CG11403

mus312
